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EXPERIMENT 04
Data modelling and form design

Aim

Design a normalized admission-enquiry data model and build forms with proper types, relationships and validation.

Procedure

1. Create forms: Student_Enquiry, Course, Follow_Up. Use stable link names. Choice lists: Last Qualification, Source Of Enquiry, Status, Mode.
2. Use lookups: Student_Enquiry.Course Interested -> Course (multi-select); Follow_Up.Enquiry -> Student_Enquiry (multi-select). One enquiry can interest many courses and one follow-up can cover many enquiries (many-to-many). A junction form (e.g. Enquiry_Course) is the stricter normalized alternative if per-pair data is needed.
3. Set mandatory fields, dropdown choices and duplicate rules (Enquiry ID and Email unique). Keep Phone Number with country code.
4. Do not delete enquiries that have follow-up history; set Status to Not Interested or Rejected instead.
5. On create, set Status to New on the server in a workflow, not only by disabling UI fields.
6. Load synthetic fixtures only (e.g. EQ001 Alice Johnson). Never import real student data.

Implementation and verification

Form	Fields and types (from your application)
Student_Enquiry	Enquiry ID (text, unique); Student Name; Email; Phone Number (with country code); Date Of Birth (date); Last Qualification (dropdown: High School, College); Percentage (decimal); Course Interested (multi-select lookup to Course); City; Source Of Enquiry (dropdown: Website, Social Media, Advertisement, Friends); Message (multi-line); Status (dropdown: New, Admitted, Not Interested, Rejected).
Course	Course Name; Duration (dropdown: 1 Year, 2 Years, 3 Years, 4 Years); Available Seats (number).
Follow_Up	Enquiry (multi-select lookup to Student_Enquiry); Follow Up Date (date); Mode (dropdown: Call, Email, Visit); Remarks (multi-line); Next Follow Up Date (date).

Fixture data in the application: 5 enquiries (EQ001 to EQ005), 5 courses (Artificial Intelligence, Digital Marketing, Engineering Physics, Business Administration, Computer Science) and 5 follow-up records.

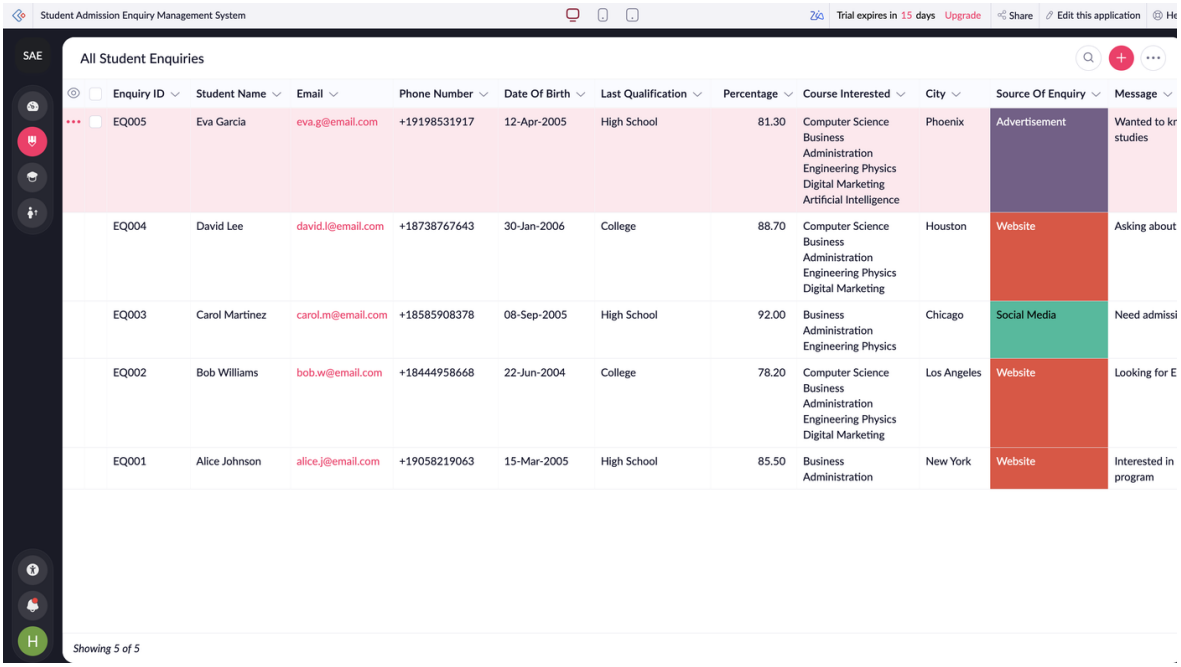
Evidence and testing

E04: relationship diagram, field dictionary, form screenshots, test records. Try a duplicate Enquiry ID, missing Email and invalid email; expect rejection. Derived values (seats remaining, enquiries per status) should be computed or updated by workflow, not left as drifting editable counts.

Test input	Expected	Actual (record)
Duplicate Enquiry ID (EQ001)	Rejected: unique rule	_____
Missing Email	Rejected: mandatory	_____
Email 'alice.j@' (invalid)	Rejected: format	_____
Valid new enquiry	Saved with Status New	_____

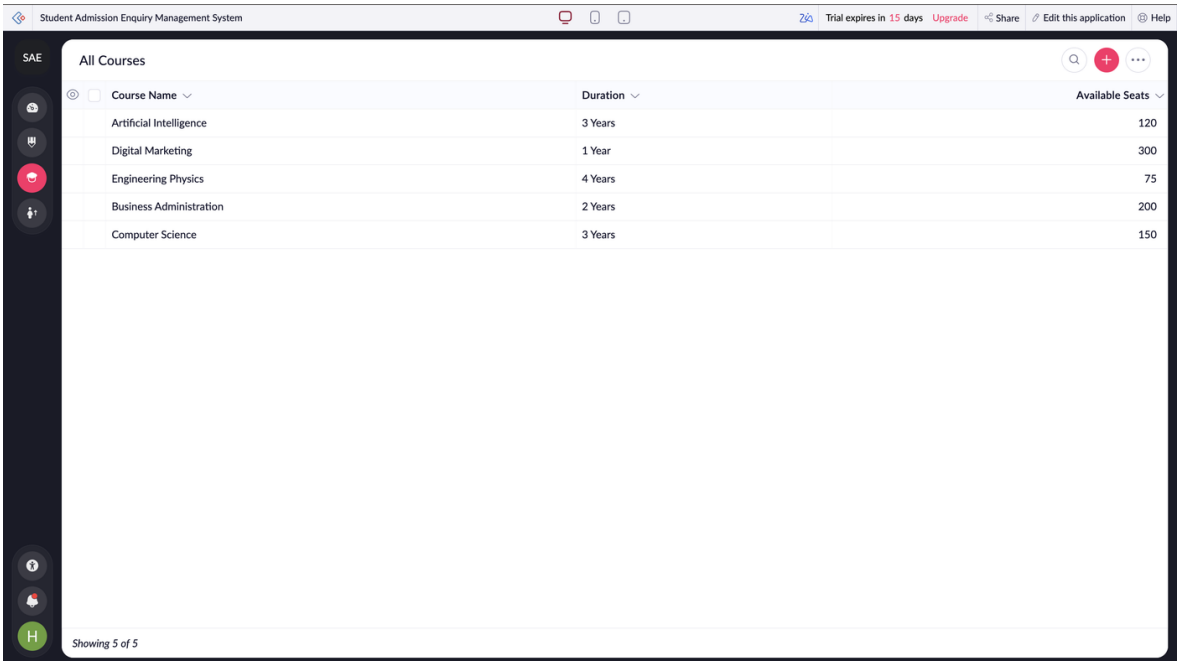
Observation from the Statuses board (Experiment 5, Figure 2): only four of the five enquiries appear in a status column. EQ001 (Alice Johnson) has no Status value. The fix is to make Status mandatory with default 'New' set by the workflow, as in step 5.

Application images and explanation



Enquiry ID	Student Name	Email	Phone Number	Date Of Birth	Last Qualification	Percentage	Course Interested	City	Source Of Enquiry	Message
EQ005	Eva Garcia	eva.g@email.com	+19198531917	12-Apr-2005	High School	81.30	Computer Science Business Administration Engineering Physics Digital Marketing Artificial Intelligence	Phoenix	Advertisement	Wanted to kr studies
EQ004	David Lee	david.l@email.com	+18738767643	30-Jan-2006	College	88.70	Computer Science Business Administration Engineering Physics Digital Marketing	Houston	Website	Asking about
EQ003	Carol Martinez	carol.m@email.com	+18585908378	08-Sep-2005	High School	92.00	Business Administration Engineering Physics	Chicago	Social Media	Need admissi
EQ002	Bob Williams	bob.w@email.com	+18444958668	22-Jun-2004	College	78.20	Computer Science Business Administration Engineering Physics Digital Marketing	Los Angeles	Website	Looking for E
EQ001	Alice Johnson	alice.j@email.com	+19058219063	15-Mar-2005	High School	85.50	Business Administration	New York	Website	Interested in program

Figure 1: All Student Enquiries report (Student_Enquiry form data)



Course Name	Duration	Available Seats
Artificial Intelligence	3 Years	120
Digital Marketing	1 Year	300
Engineering Physics	4 Years	75
Business Administration	2 Years	200
Computer Science	3 Years	150

Figure 2: All Courses report (Course form data)

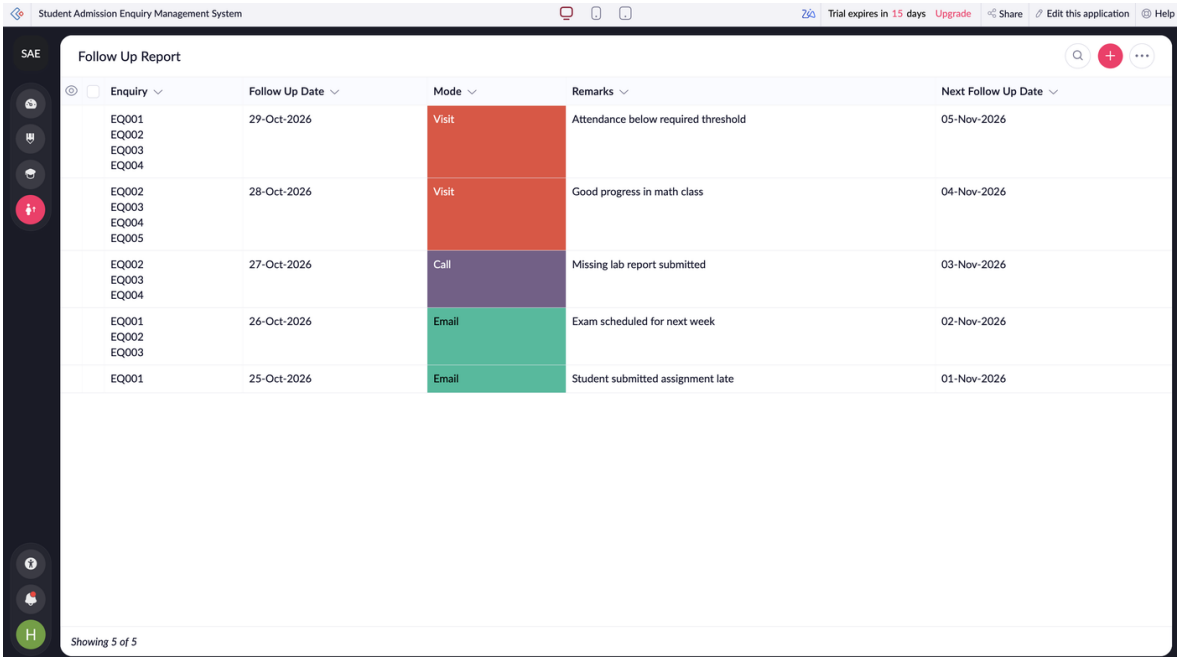
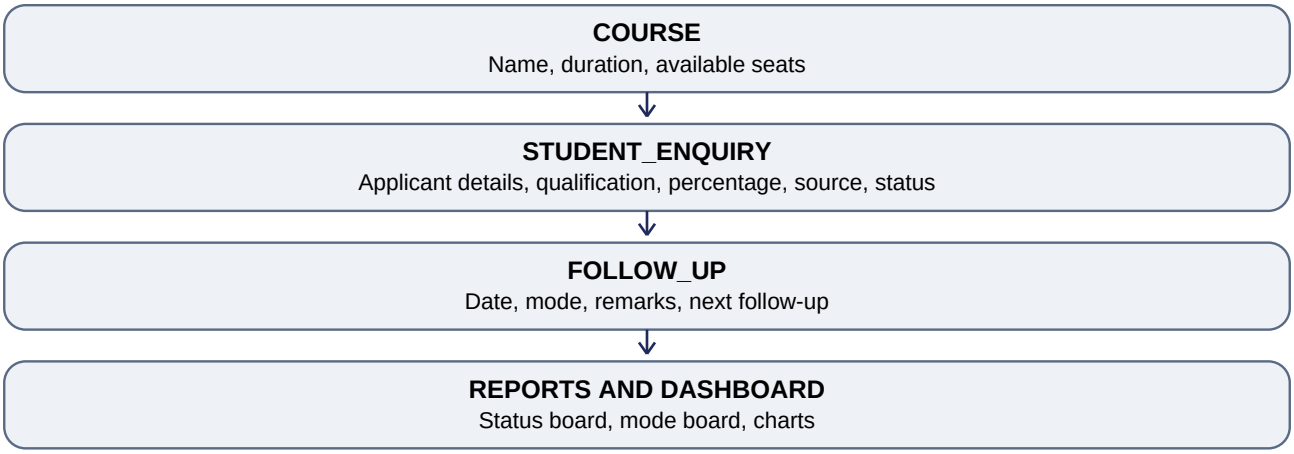


Figure 3: Follow Up Report (Follow_Up form data)

Add the form entry screens (the '+' view of each form) here if your faculty requires form screenshots; the reports above show the stored fields and relationships.

Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

Result

Expected result and verification criteria. All forms keep the intended relationships and constraints; invalid input is rejected; history is protected. No uniqueness or allocation guarantee is claimed from a lookup filter alone.